

HAT YAI INTL, Thailand

WMO: 485690

Lat: 6.933N

Lon: 100.393E

Elev: 90

StdP: 14.65

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	70.6	71.9	65.7	95.1	84.0	66.9	99.4	83.2	18.2	86.1	16.5	85.9	0.7	270	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.5	95.3	77.9	93.9	78.0	93.1	78.0	81.8	88.8	80.9	88.5	80.2	87.9	8.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.9	86.3	79.0	151.1	84.8	78.5	148.6	84.2	45.6	89.0	44.7	89.1	44.0	88.4	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	14.1	12.2	67.8	97.9	2.4	1.9	66.1	99.3	64.7	100.4	63.4	101.4	61.7	102.8		
			66.3	83.6	2.4	1.4	64.6	84.6	63.2	85.4	61.9	86.2	60.2	87.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	80.1	81.3	83.1	84.2	84.1	83.5	83.0	82.8	82.3	81.7	80.6	79.8
	DBStd	2.25	1.73	1.73	1.67	1.72	1.65	1.61	1.56	1.73	1.61	1.58	2.03	2.00
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11757	934	877	1026	1027	1057	1004	1022	1017	969	983	919	922
	CDD65	6282	469	457	561	577	592	554	557	552	519	518	469	457
	CDH74	61058	4159	4531	5871	6171	6287	5804	5734	5526	4857	4511	3929	3678
	CDH80	23919	1367	1817	2639	2819	2686	2426	2368	2270	1813	1531	1166	1017
Wind	WSAvg	3.6	4.5	4.8	4.1	3.0	3.0	3.3	3.7	3.9	3.4	3.0	3.0	3.7
Precipitation	PrecAvg	70.6	3.0	1.5	3.9	5.0	5.1	4.9	4.6	5.5	6.2	9.2	11.4	9.6
	PrecMax	102.5	14.4	6.3	13.7	11.1	12.0	9.6	8.3	12.3	9.9	17.0	27.4	25.9
	PrecMin	50.0	0.3	0.0	0.2	0.4	2.2	0.6	1.5	1.3	3.3	4.6	4.6	1.5
	PrecStd	11.2	3.3	1.7	3.1	2.6	1.9	2.2	1.8	2.3	1.9	2.9	5.5	5.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.3	93.6	96.4	97.9	96.8	94.9	94.7	95.0	93.4	92.6	91.3	90.0
		MCWB	77.1	75.9	76.5	77.6	78.4	78.3	78.0	77.5	78.0	78.3	78.5	78.0
	2%	DB	89.2	91.6	94.3	95.3	94.7	93.3	93.0	93.1	91.7	91.1	89.6	88.8
		MCWB	76.7	76.0	77.0	78.1	78.8	78.5	77.9	77.6	78.1	78.4	78.4	77.7
	5%	DB	87.9	90.0	92.9	93.6	93.0	91.8	91.6	91.5	90.6	89.5	88.0	87.6
		MCWB	76.4	76.0	77.1	78.3	79.1	78.6	78.0	77.8	78.1	78.4	78.5	77.2
	10%	DB	86.7	89.1	91.3	91.8	91.3	90.6	90.0	89.9	89.2	87.8	86.3	85.8
		MCWB	76.0	75.9	77.0	78.5	79.2	78.6	77.8	77.9	78.0	78.5	78.3	76.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	79.6	81.6	82.4	82.7	81.9	81.6	81.8	81.7	81.6	81.8	80.7
		MCDB	86.5	87.8	89.7	90.4	89.5	89.4	88.8	88.9	88.5	87.9	87.0	86.2
	2%	WB	78.9	78.3	79.6	80.8	81.1	80.7	80.0	80.1	80.1	80.1	80.2	79.2
		MCDB	85.4	86.7	89.0	89.8	89.2	88.7	87.9	88.0	87.3	86.9	85.6	84.9
	5%	WB	78.0	77.5	78.7	79.9	80.4	79.9	79.3	79.2	79.2	79.3	79.3	78.5
		MCDB	84.6	85.8	88.3	88.9	88.7	87.9	87.4	87.3	86.7	86.1	85.0	84.2
	10%	WB	77.1	76.9	78.0	79.2	79.7	79.2	78.7	78.6	78.6	78.7	78.7	77.8
		MCDB	83.8	85.1	87.0	88.1	87.8	87.2	87.0	86.7	86.1	85.4	84.4	83.1
Mean Daily Temperature Range	5% DB	MDBR	14.0	16.8	17.9	17.5	15.9	15.5	15.4	15.5	14.7	13.6	11.7	11.5
		MCDBR	15.5	18.2	20.1	19.4	17.9	17.1	17.3	17.5	16.8	15.7	14.5	14.6
		MCWBR	5.7	5.8	6.2	5.8	5.9	6.0	5.9	6.0	6.0	6.2	5.9	5.7
	5% WB	MCDBR	13.7	16.6	17.6	17.6	16.2	15.8	15.7	15.8	15.1	14.2	12.9	12.3
		MCWBR	6.2	6.1	6.4	6.4	6.3	6.4	6.1	6.4	6.3	6.3	6.1	5.9
Clear-Sky Solar Irradiance	taub	0.459	0.466	0.475	0.466	0.451	0.459	0.464	0.462	0.453	0.451	0.454	0.462	
	taud	2.289	2.247	2.246	2.279	2.346	2.350	2.326	2.316	2.340	2.349	2.339	2.301	
	Ebn at Noon	272	274	271	269	267	262	262	267	273	274	272	269	
	Edh at Noon	43	46	46	44	40	39	40	42	41	41	40	42	
All-Sky Solar Radiation	RadAvg	1543	1839	1863	1829	1639	1573	1548	1585	1545	1461	1332	1303	
	RadStd	114	107	150	86	86	71	69	56	58	100	109	136	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+137
(47) Regional (16 neighbors)	+0.40	+0.27	N/A	+0.58	+0.38	+0.34	N/A	N/A	+169	+169

Nomenclature: See separate page